Lab 02 – First Contact with OpenPLC

Industrial Security (Bachelor) • Maps to Section 02 • 30–45 min

LAB 02

Goal

Bring up the lab Docker stack and confirm that the soft PLC accepts and runs a ladder-logic program. End the lab with a Modbus read working against it.

Step 1 – Bring up the stack

```
cd bachelor/labs/_stack
docker compose up -d        # one shot: brings every service up
docker compose ps           # all services should show Up
```

The `openplc-init` sidecar uploads `conveyor.st` and starts the PLC automatically, so Modbus on port 502 is open within ∼20 s of `up -d`.

Step 2 – Open the lab landing page & the OpenPLC web UI

Browse to `http://127.0.0.1:8000` for the lab home page (overview of every lab and the stack). Then open `http://127.0.0.1:8080` for the OpenPLC web UI; log in with the default OpenPLC credentials (`openplc` / `openplc`) and change the password right away. Record the new password in your notes.

Step 3 – Inspect the running ladder program

The `openplc-init` sidecar has already uploaded `conveyor.st` and clicked **Start PLC** for you. Open the **Programs** page to see the uploaded program, then open the **Monitoring** page and watch `Motor` toggle when you flip `Start`. Note the cycle time shown at the bottom of the page.

Optional: stop the PLC, then upload `bachelor/labs/_stack/scripts/ladder/conveyor.st` yourself from your filesystem to see the workflow end-to-end – this is what the init container automated.

Step 4 – Read holding registers

```
docker compose exec student python3 /labs/scripts/modbus_read.py
```

You should see ten register values. They will all be zero unless the program has written to them. Record the first three values in your notes.

Deliverable

Hand in: (a) a screenshot of the OpenPLC *Monitoring* page with your running program, (b) the terminal output of `modbus_read.py`, and (c) the observed PLC cycle time.

Caution

The Modbus listener is bound to `127.0.0.1:502` on your host. Do not expose it to your network. The OpenPLC default credentials are well known – treat them as already compromised until you change them.